Foundation Models

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Foundation models [1] are large-scale neural networks pretrained on broad datasets and designed to provide reusable representations for many downstream tasks. In contrast to traditional specialized models, which are developed independently for individual tasks, foundation models aim to transfer knowledge across tasks and domains through prompting, probing, fine-tuning, or parameter-efficient adaptation. Their success is driven by three main properties: *scalability*, where performance often improves systematically with model size, data, and computation; *adaptability*, where the same pretrained backbone can support multiple downstream applications; and *reuse*, where high-quality pretrained representations reduce the amount of task-specific data and model development required. These advantages come with important challenges, including high computational and energy costs, robustness under distribution shift, fairness and bias, interpretability, and the difficulty of adapting general-purpose models to specialized domains such as medical imaging.

1 Architectural Foundations

Modern vision foundation models build on several important architectural developments. Vision Transformers (ViTs) [2] represent images as sequences of patches and use transformer self-attention instead of the strong locality and translation biases of conventional convolutional networks. Their ability to benefit from large-scale training established transformers as powerful general-purpose visual backbones. Swin Transformers [3] introduced hierarchical representations and shifted-window attention, reducing the computational cost of global attention while producing multiscale features suitable for dense prediction. ConvNeXt [4, 5] subsequently showed that carefully modernized convolutional networks can incorporate many successful transformer-era design principles while retaining a convolutional architecture. These developments are closely connected to empirical scaling laws [6], which describe systematic relationships between model capacity, training data, computation, and performance. Later work [7] has emphasized that effective scaling requires balancing model size and data rather than increasing model capacity alone, motivating the increasingly large but carefully trained backbones used in contemporary foundation models.

2 Self-Supervised Learning

Self-supervised learning (SSL) [8] enables models to learn transferable representations from unlabeled data and has become a central component of visual foundation-model pretraining. Contrastive approaches such as SimCLR [9, 10] and MoCo [11, 12, 13] learn invariance by bringing representations of different augmentations of the same image together while separating unrelated samples. Non-contrastive and self-distillation methods such as BYOL [14], VICReg [15], and DINO [16] remove the need for explicit negative pairs through architectural asymmetry, teacher-student learning, or regularization against representational collapse. Masked prediction provides another major direction: MAE [17] reconstructs masked image content, while methods such as iBOT [18] combine masked prediction with self-distillation at the patch level. More recent approaches, including DINOv2 [19], scale these ideas using large curated datasets, image- and patch-level objectives, distillation, and improved regularization to obtain representations that transfer to both image-level and

dense prediction tasks. Predictive approaches such as I-JEPA [20] further shift the objective from reconstructing pixels toward predicting latent representations of missing visual content.

3 Task-Specific Foundation Models

Foundation-model principles have also been extended from general-purpose visual representation learning to task-oriented models. The Segment Anything Model (SAM) [21] introduced promptable segmentation using a large pretrained image encoder, a prompt encoder, and a lightweight mask decoder. Given prompts such as points or bounding boxes, SAM can produce object masks without retraining for each segmentation problem. SAM 2 [22] extends this idea to both images and videos through a memory-based architecture that propagates information across frames. These models illustrate how large-scale pretraining can transform a traditionally task-specific problem into a reusable and interactive system. At the same time, performance can degrade when such models are applied to domains that differ substantially from their pretraining distribution, motivating domain-specific adaptation strategies such as SAM adapters and medical variants of SAM.

4 Finetuning and Parameter-Efficient Adaptation

Although foundation models provide broadly useful representations, most specialized applications still require some form of adaptation. Full fine-tuning updates all model parameters and can provide strong task-specific performance, but becomes increasingly expensive as model size grows and may alter previously learned capabilities. In-context learning [23] offers a different form of adaptation in language models, where examples supplied in the prompt influence predictions during the forward pass without changing model parameters. Parameter-efficient fine-tuning (PEFT) [24] instead keeps most pretrained weights frozen and updates only a small subset of parameters. Common approaches include selective fine-tuning, learnable prompts or prefixes, bottleneck adapters [25], and low-rank reparameterization such as LoRA [26, 27, 28]. The effectiveness of these approaches is related to evidence that adaptation of highly overparameterized pretrained models can often occur within a much lower-dimensional optimization subspace.

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